

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T17 · Algebraic Roots & Indices

Questions

Student book · questions with writing space

■ 43 pages

FAMILY MAP

Algebraic Roots Indices

F01

Combine indexed algebraic factors

24 Q | 48 marks

F02

Interpret zero, negative and fractional indices

11 Q | 23 marks

F03

Solve an index relation by a common base

7 Q | 18 marks

Working Unit 02

4MA1-1H Q4

UNIT 1
2 marks

4 $x^5 \times x^7 = x^m$

(a) Find the value of m

$m = \dots\dots\dots$
(1)

$y^8 \div y^3 = y^n$

(b) Find the value of n

$n = \dots\dots\dots$
(1)

Working Unit 03

4MA1-1H Q4

UNIT 1

2 marks

(c) Simplify fully $(5a^4r^2)^3$

(2)

4MA1-2H Q15

UNIT 1

(Total for Question 15 is 3 marks)

Question 06

4MA1-1H Q10

MODULAR UNIT 1

4 marks

10 (a) Simplify $(2p)^0$ where $p > 0$

.....
(1)

$$y^9 \times y^{-3} = y^n$$

(b) Find the value of n

$n =$
(1)

(c) Simplify fully $(5a^4c^2)^3$

.....
(2)

Question 27

4MA1-2H Q23

MODULAR UNIT 1

3 marks

23 Simplify $\frac{30 \times 25^{2x+7}}{\sqrt{180} \times (\sqrt{5})^{4x+9}}$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.

Question 17

4MA1-2H Q4 (b)

MODULAR UNIT 1

1 marks

(b) Simplify $h^{15} \div h^3$

.....

(1)

Question 18

4MA1-2H Q4 (c)

MODULAR UNIT 1

2 marks

(c) Simplify fully $(2g^3k^5)^4$
(2)

4MA1-2H Q4 (d)

2 marks

$$n = \dots\dots\dots (2)$$

Question 25

4MA1-2H Q12 (b)

MODULAR UNIT 1

2 marks

(b) Simplify fully $(125p^6y^{24})^{\frac{2}{3}}$

.....
(2)

Question 02

4MA1-1H Q6

MODULAR UNIT 1

2 marks

6 (a) Simplify $(2c^4d^7)^3$

(2)

Question 03

4MA1-1H Q6

MODULAR UNIT 1

1 marks

(b) Find the value of $5y^0$ where $y > 0$

.....

(1)

Question 19

4MA1-2H Q15

MODULAR UNIT 1

2 marks

15 (a) Simplify fully $(32a^{15})^{\frac{3}{5}}$

.....
(2)

Question 20

4MA1-2H Q15

MODULAR UNIT 1

2 marks

(b) Express $\left(\frac{1}{10x}\right)^{-3}$ in the form px^n where p and n are integers.

(2)

Question 01

4MA1-1H Q3 M01

MODULAR UNIT 1

1 marks

3 (a) Simplify $(p^3)^5$

(1)

Question 23

4MA1-2H Q14 M01

MODULAR UNIT 1

3 marks

14 Given that $\frac{3^{2n+3}}{3^4} = 3^3 \times 3^{1-2n}$

find the value of n

Show your working clearly.

$n =$

Question 04

4MA1-1H Q9 Q9(a)

MODULAR UNIT 1

1 marks

9 (a) Simplify $8 \times (4t)^0$

(1)

Question 05

4MA1-1H Q9 Q9(b)

MODULAR UNIT 1

1 marks

$$x^6 \div x^{-5} = x^p$$

(b) Find the value of p $p = \dots\dots\dots$

(1)

Question 06

4MA1-1H Q9 Q9(c)

MODULAR UNIT 1

2 marks

(c) Simplify fully $(2k^2m^4)^3$

(2)

Question 29

4MA1-2H Q24 Q24

MODULAR UNIT 1

3 marks

24

$$\frac{18 \times (\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

Express x in terms of n

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$

Question 01

4MA1-1H Q1 Q1(a)

MODULAR UNIT 1

1 marks

1 (a) Simplify $a^7 \times a^4$

(1)

Question 02

4MA1-1H Q1 Q1(b)

MODULAR UNIT 1

1 marks

(b) Simplify $w^{15} \div w^3$

(1)

Question 03

4MA1-1H Q1 Q1(c)

MODULAR UNIT 1

2 marks

(c) Simplify $(8x^5y^3)^2$

(2)

Question 11

4MA1-1H Q13 Q13(b)

MODULAR UNIT 1

3 marks

(b) Simplify completely $\left(\frac{16w^8}{y^{20}}\right)^{-\frac{3}{4}}$

(3)

(c) Simplify $(p^2 + 3)^0$

(b) Simplify fully $\left(\frac{27a^{12}}{t^{15}}\right)^{-\frac{2}{3}}$

1 (a) Simplify $\frac{x^9}{x^2}$

(b) Write $\frac{7^8 \times 7^4}{7^3}$ as a single power of 7

9 (a) Simplify $(2x^3y^5)^4$

(b) work out the value of n .

(1)

11 Simplify fully $\left(\frac{9t^4w^9}{18t^6w^{10}}\right)^{-}$

(c) work out the value of n .

(a) Find the value of x .
Show your working clearly.

$$x = \dots\dots\dots (2)$$

(b) Find the value of n .
Show your working clearly.

(b) Find the value of n .
Show your working clearly.

14 Simplify fully $\left(\frac{9x^4}{16y^{10}}\right)^2$

8 (a) Simplify $(3x^2y)^0$

12 (a) Simplify $(64p^9q^{12})^{\frac{2}{3}}$

12 Given that $\frac{3^x}{9^{3x}} = 81$

find the value of x .

Show clear algebraic working.

$x =$

(Total for Question 12 is 3 marks)

(a) Work out the value of m

$$m = \dots\dots\dots (1)$$

(b) Work out the value of n

$$n = \dots\dots\dots (1)$$

(Total for Question 12 is 2 marks)

$$\frac{10^{4n} \times 2^{3(n^2-5n)} \times 5^{2(1-2n)}}{20^2} = 1$$

8 $\frac{8^{-2} \times 8^9}{8^{10}} = 8^n$

Work out the value of n

$n = \dots\dots\dots$

(Total for Question 8 is 2 marks)

13 (a) Simplify fully $\left(\frac{125x^6}{y^{15}}\right)^{\frac{2}{3}}$

.....
(2)

Given that $4^n = \frac{4^m}{64^p}$

(b) Express n in terms of m and p

$n =$
(2)

(Total for Question 13 is 4 marks)